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Microfibers in laundry wastewater: problem and solution

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Abstract

Data corroborated in this study highlights laundry wastewater as a primary source of microfibers (MFs) in the aquatic environment. MFs can negatively impact the aquatic ecosystem via five possible pathways, namely, acting as carriers of other contaminants, physical damage to digestive systems of aquatic organisms, blocking the digestive tract, releasing toxic chemicals, and harbouring invasive and noxious plankton and bacteria. This review shows that small devices to capture MFs during household laundry activities are simple to use and affordable at household level in developed countries. However, these low cost and small devices are unreliable and can only achieve up to 40% MF removal efficiency. In line filtration devices can achieve higher removal efficiency under well maintained condition but their performance is still limited compared to over 98% MF removal by large scale centralized wastewater treatment. These results infer that effort to increase sanitation coverage to ensure adequate wastewater treatment prior to environmental discharge is likely to be more cost effective than those small devices for capturing MFs. This review also shows that natural fabrics would entail significantly less environmental consequences than synthetic materials. Contribution from the fashion industry to increase the share of natural fabrics in the current textile market can also reduce the loading of plastic MFs in the environment.

Keywords: Laundry activities, Microplastic fibers, Natural fabrics, Washing machines, Wastewater treatment.

1. Introduction

Microfibers (MFs) pollution is an emerging environmental problem of a global scale.

Specific and widespread evidence has emerged in recent years to show a range of harmful effects from MFs on aquatic habitats and even potential impacts on human health (Bilal et al., 2020). Scientists estimated that millions of tonnes of MFs enter the aquatic environment each year from the textile industry and laundry household (Napper and Thompson, 2016; Carney Almroth et al., 2018). MFs are particularly mobile, thus, even after landfilling, they can be transferred to leachate and eventually find their way to the aquatic environment (Silva et al., 2021). Synthetic MFs such as polyester, acrylic, and nylon are non-biodegradable, thus, once entered the aquatic environment, they are present everywhere from shorelines (Barrows et al., 2018) to the ocean. MFs have been detected in the digestive system of aquatic organisms, then they accumulate in the food chain, especially in fish and shellfish (Waring et al., 2018; Qiao et al., 2019; Weis, 2020; Jong et al., 2022; Zhang et al., 2022). MFs can also release toxic chemicals such as plasticizers that have been added to textiles during manufacturing to modify their physical properties (Sridharan et al., 2022). MFs can act as transport vectors to carry these chemicals to affected organisms in the environment. Humans are also indirectly exposed to microfibers through the food and drinking water.

Recent demand and consumption of synthetic fabrics have further exacerbated the problem of MFs in the environment. Indeed, the demand for synthetic fabrics has grown exponentially over the last decade. Between 2010 to 2019, the compound annual growth rate (CAGR) of the global synthetic fabric market was 4%. It is forecasted to grow to over 7%/yr over the next decade (Zhang and Sesto, 2019). It is also estimated that Asia will experience a faster CAGR growth due to significant economic development in the region.

Synthetic fabrics currently account for 63% of the worldwide fabric market, the majority of which is polyester. Polyester is the predominant source (52%) of synthetic fabric, followed

by polyamide (5%), and other synthetic (6%) ([Anonymous, 2020](#)). Polyester fabric has been widely used for multipurpose (e.g., plastic packaging and fashion) due to its mechanical strength, aesthetic appearance, lightweight, and low cost. Polyester accounted for more than half of global MF textile in 2018 (around 55 million tonnes) from the textile industry ([Freitas et al., 2017](#)).

Synthetic fabrics such as nylon, polyester, spandex, and acrylic were initially produced as low-cost alternatives to natural fabrics as they can be synthesized from petroleum-based chemicals ([Franco et al., 2020](#)). For example, acrylic is used to imitate wool due to its good heat insulation properties ([Ahmad et al., 2020](#)). Artificial fur and fleece made from acrylic fibers significantly reduce the materials cost for cloth manufacturing. Early synthetic fabrics had inferior texture properties (e.g., look and feel) compared to natural fabrics. Over time, progress in materials science and engineering has significantly improved the texture properties of synthetic fabrics, making them similar and, in some cases, even better than natural materials ([Egan et al., 2022](#)). As a result, synthetic fabrics have gradually displaced natural fabrics to dominate the textile industry.

As the potentially significant and irreversible impact of the presence of MFs on the environment has become clearer recent years, dedicated scientific studies and engineering measures have also emerged to limit their release to the natural environment. The release of MFs to the natural environment is closely linked to laundry activities and the textile industry through wastewater discharge ([Henry et al., 2019](#)).

This work aims to critically examine laundry wastewater and from the textile industry to seek a holistic solution to the issue of MFs in the natural environment. Potential effects of MFs on the ecosystem are highlighted to provide a context to this review. Data corroborated from the literature are then analysed to delineate key factors (e.g. temperature, fabric types, detergents,

and washing methods) influencing the release of MFs to wastewater during laundry activities. Current and future approaches to reduce or eliminate the release of MFs to the natural environment are then explored and discussed.

2. Microfibers in laundry wastewater

2.1. Microfibers

As a subset of microplastics, MFs are small plastic particles of less than 5 mm in size (Lebreton et al., 2018; Bui et al., 2020; Xu et al., 2020). Some MFs from textiles are longer than 5 mm, but their diameter fits into microplastics categories. In addition, after entering the environment, these long plastic fibers can also be broken down into segments less than 5 mm in length (De Falco et al., 2020). MFs are essentially primary microplastics released into the environment along the supply chain of the fashion industry from textile manufacturing, fashion production, use and laundry of textile products, and textile waste disposal (Henry et al., 2019).

Previous studies (Booth et al., 2017; Uddin et al., 2020) have confirmed significant annual input of microplastics to the environment, mainly from poorly managed waste and wastewater disposal. Due to their highly mobile characteristics, in the aquatic environment, MFs are arguably the most important subgroup of microplastics (Kazmiruk et al., 2018; Yang et al., 2019).

Human activities are the main source of MFs in the aquatic environment (Belzagui et al., 2019). Annual release of MFs to the environment via wastewater discharge varies from country to country ranging from hundreds to tens of thousands tonnes/yr (Table 1). China accounts for more than 40% (about 70,000 tonnes) of the global MFs emissions to the aquatic environment, mostly through untreated wastewater discharge (Gavigan et al., 2020). North

America also contributes approximately 880 tonnes/yr (Vassilenko et al., 2019). At the other end of the spectrum, Finland contributes 150 tonnes/yr of MFs to the environment.

[TABLE 1]

Data in Table 1 suggests that MFs release to the environment is closely related to the textile industry and laundry wastewater. China, India, and Southeast Asian Nations (ASEAN) have the largest textile industry globally. These countries are responsible for approximately 86% of the total MFs released from the fashion industry supply chain (Freitas et al., 2017; Henry et al., 2019). According to Lares et al. (2018), around 97% of MFs is from polyester fabric, followed by 3% polyamide of total MFs. In addition, it was estimated that about 7x10⁵ MFs were emitted per washing of 6 kg of clothes in each household (Napper and Thompson, 2016). Data from the literature conclusively points to laundry activities as the largest source of MFs to the environment (Yang et al., 2019).

2.2. Environmental effects of MFs

In the aquatic environment, MFs can cause several environmental consequences (Fig. 1). They can act as carriers of harmful contaminants, algae species and pathogens (Kwon et al., 2017; Amelia et al., 2021; Allouzi et al., 2021) from one place to another in the aquatic environment. Due to the high absorptive capacity for hydrophobic organic chemicals and a large surface area to the volume ratio, co-contaminants could attach to MFs (Amelia et al., 2021) to produce toxic aggregates. In this case, MFs become a vector or vehicle for transferring toxic co-contaminants to organisms via a direct food uptake way (Fig. 1). Heavy metals (e.g., Cd and Pb) and toxic organics such as Chlordecone, bisphenol A, phthalates, and polybrominated diphenyl ethers (Scopetani et al., 2018; Weis, 2020; Yan et al., 2020; Cormier et al., 2021) have been detected on the surface of MFs in aquatic environment.

Previous studies indicated that toxic chemicals could sorb to MPs, particularly the surface of aging MPs, and could negatively impact the growth of some aquatic organisms (Cormier et al., 2021). For example, tributyltin is a toxic organic chemical and could accumulate in the food chain at a low concentration (Leung et al., 2001). The reported LC50 concentration of tributyltin could range from 0.015 ppb for *Tigriopus japonicas* (Kwok and Leung, 2005) to 36 ppb for *Tisbe biminiensis* (Costa et al., 2014). Besides, the plastic surface could also be a reservoir of pathogenic bacteria (e.g., *Aeromonas salmonicida*) through biofilm (Viršek et al., 2017; Rotjan et al., 2019; Sooriyakumar et al., 2022). Due to the protection of the biofilm, some microorganisms living in the biofilm are tolerant to toxic organic chemicals, such as antibiotics and antimicrobial agents, as well as heavy metals. Then, the biofilm becomes thicker and separates from the microplastic surface, contributing to the transfer of those toxic substances from one place to another in the aquatic environment (Bolan et al., 2020; Sooriyakumar et al., 2022).

Aquatic organisms digest MPs through their gills or by confusion with their prey (Savoca et al., 2019; Zantis et al., 2021). Among sub-groups of MPs, MFs was the most prevalent in the digestive system of an aquatic animal at about 60%, followed by plastic fragment (26%), films and pellets (14%) (Issac and Kandasubramanian, 2021). MFs could cause more severe damage to digestive system than microbeads and large MP fragments because of their high mobility. Once entered the digestive system, MFs can form entanglement causing intestinal blockage and other damages (Qiao et al., 2019; Weis, 2020) (Fig.1). In addition, MFs could decrease the mucus volume in the gut, leading to the decreasing working capacity immune function of the intestinal epithelium and intestinal epithelial damage (Qiao et al., 2019).

Inside the animal body, MPs can release co-contaminants, plasticizers, and other chemicals that can bioaccumulate in important organs, especially liver (Fig. 1). Barboza et al. (2018) observed that mercury concentration in the liver was higher than that in the gills of *D. labrax*

juveniles. The level of Hg in gills ranged from 1.6 to 5.2 µg/g, whereas in the liver, they ranged from 3 µg/g to 8 µg/g. The toxic element Cd, absorbed into MPs, was mainly detected in the liver (Lu et al., 2018). Ag accumulated in the intestinal mucous, mucosal epithelium, muscle layer, and serosa (Khan et al., 2017). The accumulation of toxins in the liver and tissues could seriously impact organisms, such as DNA damage, oxidative stress, neurotoxicity, and physiological effects.

In the environment, MPs can harbor and encourage the proliferation of invasive phytoplankton and noxious species (Fig. 1), resulting in biodiversity loss and posing a threat to the aquatic ecosystem (Shahul Hamid et al., 2018; Thushari and Senevirathna, 2020). Some alien species often used the surface of floating MPs in the aquatic environment as a habitat for growth. About 2.5×10^{11} particles of MPs buoyant in the Mediterranean Sea (Werner et al., 2016) were potential carriers of the invasive species based on MPs' biofouling and persistent properties. Casabianca et al. (2019) provided evidence that toxic microalgal species (e.g., *Bacillariophyceae* and *Dinophyceae*) could survive on the surface of floating MPs in high amounts ranging from 61-100 cells/mg in the Mediterranean Sea. With the change of ocean currents and sea wind, MPs - a toxic vector could be dispersed to new distant aquatic areas, subsequently increasing the toxin in marine environments due to the proliferation of invasive and noxious organisms.

[FIGURE 1]

2.3. Microfibers from laundry activities

2.2.1 Microfibers from laundry activities

Laundry activities have been identified as the most significant source of MFs in the environment (Hernandez et al., 2017; De Falco et al., 2019). MFs can enter the environment throughout the supply chain of the fashion industry from textile fabrication, clothing manufacturing to including domestic laundry activities (Acharya et al., 2021; Ross et al.,

2021). The release of MFs from textile fabrication and clothing manufacturing can be readily addressed via centralized wastewater treatment (Bui et al., 2020; Pedrotti et al., 2020). On the other hand, the dispersed nature of domestic laundry activities makes them especially difficult to control.

2.2.2 Factors influencing the release of microfibers from laundry activities

The release of MFs from the laundry depends on several factors, such as textile characteristics (e.g., fabric age and type), detergent properties and strength, washing temperature, and abrasion during washing (Fig. 2). New clothes shed more MFs during washing due to residues from the fabric production process (Belzagui et al., 2019; De Falco et al., 2019). In addition, the first few washing cycles also release significantly more MFs than later cycles when the release of MFs have reached a steady state. The release of polyester MFs was only 1/3 in the second wash compared to the first time of new clothes, and the number of shedding MFs dropped by 1/10 on the fifth wash for all polyester and cotton MFs (Sillanpää and Sainio, 2017). MFs shedding also depends on fabric materials. MFs shedding from polyester fleece fabric was twice as much as acryl fabric (Kärkkäinen and Sillanpää, 2021).

The use of detergent, both in liquid and powder form, increased MFs release (Fig. 2). Detergent chemically removed dirt and debris trapped in fabrics and washed them away by generating a chemical reaction with water and producing alkalinity. However, a high pH condition (e.g., above pH 9.0 for liquid) could induce chemical damage on polyester fabrics through hydrolysis (Bishop, 1995) and thus, washing with detergents resulting in the higher release of MFs by 1.2-6.0 folds compared to without detergents (Yang et al., 2019). It is also suggested that powder detergents may result in more MFs release than liquid detergents (Issac and Kandasubramanian, 2021; Periyasamy 2021). This is because powder detergents contain insoluble particles to increase abrasion and cause friction with the fabrics.

The increase in washing temperature also resulted in an increase in MFs release (Yang et al., 2019; Zambrano et al., 2021) (Fig. 2). Hot water washing at 60°C released more polyester MFs (16×10^3 of MFs) than at lower washing temperature ($\sim 30^\circ\text{C}$) (Yang et al., 2019). Indeed, washing clothes at a lower temperature could reduce the number of released MFs by up to 50% (Cotton et al., 2020). Washing at a high temperature promotes the expansion of yarns, the bonds between the fibers become loose, creating more free spaces in the fabric structure for the mobilization of the broken MFs, particularly under high alkalinity conditions (Nishimura and Sarko, 1987). Therefore, synthetic fabric textiles should be washed at low temperatures such as 30°C , as higher temperatures promote MFs release.

[FIGURE 2]

Abrasion during laundry can contribute more to fabric damage than wear and tear, thus, releasing more MFs (Fig. 2). The surface abrasion of the fabric increases mechanical stress, breaks MFs structure, and releases more MFs during laundry activity (Jönsson et al., 2018). Besides, the abrasion depended on several factors, such as the spin-dry rate of the washing machine, the use of detergent or surfactants to remove dirt on clothes, and the loading drum. During dry spinning, the fabric can be subject to abrasion and centrifugal force, causing a further release of MFs to laundry wastewater (Sillanpää and Sainio, 2017). In addition, the high alkalinity of detergents could also produce MFs fuzz on the fabric surface during the laundry. Then, the abrasion process inside the washing machine immediately broke the fuzz, releasing MFs to the aquatic environment. According to Cai et al. (2021), the number of MFs detached from fleece after abrasion experiments were about 3,000 MFs/gram of textile fabric, which were 5-folds higher than that of non-abrasive condition.

3. Short term measures

3.1. Traps and filters

There have been several commercial products or small devices for capturing MFs during laundry activities. These products can be placed either directly inside the washing drum or fitted to washing machines to remove MFs from laundry wastewater. Examples of the former include laundry bags (e.g. Guppyfriend) and capturing balls (e.g. Cora). The latter are in-line filters such as Lint LUV-R and Filtrol™ that can be fitted directly to laundry equipment (Table 2).

[TABLE 2]

Washing bags such as Guppyfriend work by capturing MFs within the bag, and thus, reduces the number of MFs released to the aquatic environment. Other commercial products such as Cora balls have spiky tentacles to capture MFs as it moves around inside the washing drum (Kärkkäinen and Sillanpää, 2021). Although the Guppyfriend bags and Cora balls are easy to use, there is uncertainty about their MFs removal efficiency (Table 2). MF capturing efficiency by the Cora balls depends on fiber dimensions, such as length and width. The longer the fibers are, the easier they are to get caught and tend to wrap together inside the ball.

Compared to in-drum products such as Guppyfriend bags and Cora balls, in-line filters are more efficient at capturing MFs from laundry wastewater (Table 2). Most of these filters (e.g., Lint LUV-R, Filtrol™) use a stainless-steel mesh filter with a pore size of approximately 150 µm for MFs removal by straining. Since MFs are a few millimeters in length, these in-line filters can offer high and consistent removal well > 80% (McIlwraith et al., 2019).

Commercial products in Table 2 tend to have low and unreliable MF removal performance (Table 2). They must be regularly maintained and serviced. MFs must be manually removed from the Guppyfriend bags and Cora balls after every wash. Incorrect or lack of maintenance of the in-line filters can cause blockage and significantly reduce their MFs removal efficiency. Careful disposal of the trapped MFs is also required to ensure that they do not end up in the environment.

3.2. Wastewater treatment

3.2.1 The removal of microplastics by municipal wastewater treatment

In most wealthy countries, laundry water is released to the sewer for treatment at a centralized wastewater treatment plant (WWTPs). WWTPs can achieve complete or near complete removal of MFs from household laundry activities through several and often redundant treatment stages (Sun et al., 2019; Liu et al., 2021).

As denoted in Fig. 3, primary treatment accounts for the initial 40-65% removal of the MFs (Sun et al., 2019). Primary treatment (which is the first treatment stage), typically includes screening, grit and grease removal, and primary settlement. MFs are usually long, light, and easy to float on the water, so they are effectively captured by screening and floatation for oil and grease removal (Talvitie et al., 2017). Secondary treatment can add about 20-60% MF removal on top of the efficiency of primary treatment. In this stage, small and suspended MFs that have escaped primary treatment can be captured by adsorption to activated sludge flocs and then removed by the secondary clarification tank. Tertiary treatment processes such as sand filtration and microfiltration can then remove any remaining MFs in the water by size exclusion prior to effluent discharge. In this phase, an alternative technique - membrane bioreactor could reach an excellent removal performance (90–99%) compared to others, e.g., disc filter (80–98%), granular filter (80–98%), dissolved air flotation (60–85%) (Talvitie et

al., 2017; Enfrin et al., 2020). The three stages offer complementary and redundant treatment capability to ensure a high MF removal rate.

It is noteworthy that not all WWTPs include all three treatment stages. Tertiary treatment is only used when the effluent is intended for water reuse or discharged to a sensitive environment (Bui et al., 2020). Even without tertiary treatment, WWTPs can provide an effective and reliable performance for removing MFs from wastewater.

[FIGURE 3]

4. Addressing the root cause

4.1. Synthetic fabrics

Fabrics used in the fashion industry can be of natural or synthetic origin. Natural fabrics are from plant-based (e.g., cotton, linen, and jute) or animal-based (e.g., wool and silk) fibers (Syduzzaman et al., 2020). Silk is naturally obtained from an insect and produced from the substance used to form a cocoon. Wool is from sheep or other furry animals. It is a highly desirable product due to its capacity to hold heat and natural appearance. Cotton is the most common form of plant-based fibers that can be obtained from cotton seeds. Linen is a strong and lightweight fabric made from fibers of the flax plant. Jute fibers are from the skin of the jute plant. It is used to make several fabrics such as hessian cloth, sacking, scrim, carpet backing cloth, and canvas. Jute fabrics have exceptionally high strength, but they have a rough texture. Thus, their applications in the fashion industry are limited. Aside from the commonly used natural fibers (e.g., cotton and silk), regenerated cellulosic fibers (e.g., viscose, rayon) have also become popular. They are made from dissolved wood pulp. Regenerated cellulosic fibers have physical and chemical properties similar to cotton, such as water absorbency and close to skin comfort. Primary applications for regenerated cellulosic

fibers are in the hygiene and wipes sectors ([Ahmad et al., 2014](#); [Muthu, 2017](#); [Subash and Muthiah, 2021](#)).

[FIGURE 4]

The remaining is from natural fibers, mostly cotton (25%) and regenerated cellulosic fabrics (7%) with a small contribution from silk and wool. As plant-based materials, regenerated cellulosic fabrics can make the fashion industry more sustainable since the raw materials are renewable and the final products can be naturally assimilated into the environment at the end of their service life ([Anonymous, 2020](#)).

Despite significant improvement in texture properties, synthetic fabrics have an inherent problem. They are extremely persistent in the environment. The degradation of synthetic fabrics could be governed by polymer types, density, molecular size, additives, biological activity, and environmental conditions. For example, polyester (the most common synthetic fabric) is essentially polyethylene terephthalate or PET. Like a PET water bottle, it takes more than 200 years to fully biodegrade polyester in the natural environment ([Fig. 4](#)). [Royer et al. \(2021\)](#) simulated marine conditions to compare the biodegradability of regenerated cellulosic and polyester fibers. After 90 days, [Royer et al. \(2021\)](#) observed some or complete degradation of regenerated cellulosic fibers. By contrast, polyester fibers did not show any evidence of biodegradation. Compared to plant-based fabrics, animal-based materials such as wool and silk biodegradable slower, but they would be fully biodegraded in the natural environment after two to four years at the most. Overall, the persistency of synthetic fibers is hundreds to several thousand times higher than natural products ([Fig. 4](#)). Thus, replacing synthetic fabrics with natural products is the most effective strategy to reduce the impact of MF on the aquatic environment.

Natural fabrics are expensive and often unaffordable for customers in low-income countries. Consequently, developing and applying of advanced technologies in natural fabrics production could increase productivity and reduce this material's cost. These are valuable solutions to help natural fabrics reach a more comprehensive range of customers. Blending or surface coating natural fabrics with other materials could be an idea to increase fabric strength to overcome the current drawback of these materials. Besides, changing to regenerated cellulosic fabrics (rayon) was an alternative choice for attracting consumers and more safe for the environment. Rayon is a fabric made from purified cellulose fibers, which are typically created from wood pulp ([Mendes et al., 2021](#)). For example, Tencel®, a type of rayon, like viscose and modal, was also made from wood pulp or bamboo. As a plant-derived fiber, Tencel is also biodegradable. Tencel requires a lot less dye than cotton. When produced, it is also pure white, so no bleaching is necessary, and undyed is always an excellent alternative to synthetic fabrics ([Basit et al., 2019](#)).

Additional research to improve the texture properties and durabilities of natural fabrics can help them compete with synthetic materials. These can include work to improve the strength and abrasion resistance, breathability, water resistance, quick-drying properties of natural fabrics. Increasing consumer awareness about the harmful effects of synthetic fabrics can also help the effort to phase out synthetic fabrics.

4.2. Future roadmap

The ecological impact of MFs in natural water is an environmental problem of global scale. Thus, solving this problem is the responsibility of more than just individual households or the consumers. Indeed, a holistic solution should cover the entire supply chain, starting from the textile industry and including municipal wastewater treatment, washing machine manufacturers and the consumers.

The textile industry has a significant role to play in resolving this problem. By limiting the use of synthetic plastic-based fabrics, the reduction of MFs release can be achieved at the source. Natural fibers have a very short lifetime in the environment as they are fully biodegradable (Fig. 4). Although synthetic fabrics are often cheaper and easier to manufacture than natural fabrics, consumer sentiment to support nature-based products is growing in recent years. Further progress in materials science and engineering is expected to make natural fabrics more durable and with better texture properties to compete with synthetic materials.

Large-scale removal of microfibers from the environment is neither technically feasible nor economically viable, so the focus needs to be on preventing the release of MFs to the environment at the first place. Less than 50% of the global population is currently served by WWTPs. Increasing the global coverage of centralised wastewater treatment can significantly reduce MFs release to the environment. As discussed in section 2.2.2, washing condition can have some impact on the release of MFs. By reducing washing temperature, detergent consumption, and abrasion, the release of MFs to laundry wastewater can be reduced. These measures can also reduce the energy and chemical footprint of laundry activities. Last but not least, the use of MFs traps and filters should be encouraged. Although the efficiency of these devices is low and variable, they are low cost and can be readily implemented.

5. Conclusions

The release of microfibers (MFs) into the aquatic environment primarily from laundry activities has emerged as a globally significant environmental issue. There are several ways in which MFs can negatively impact the aquatic ecosystem. MFs can act as carriers of other contaminants, cause physical damage to digestive systems of aquatic organisms, block the digestive tract, release toxic chemicals to internal organs, and harbour invasive and noxious plankton and bacteria in the aquatic ecosystem. Results corroborated from this review show

that no single solution can adequately address the problem of MFs in the aquatic environment. Small devices to capture MFs during household laundry activities are low cost and simple to use. However, these devices are unreliable and can only achieve low or moderate MF removal efficiency compared to conventional wastewater treatment, which can completely intercept and prevent the release of MFs into the aquatic environment. In fact, effort to increase sanitation coverage to ensure adequate wastewater treatment prior to environmental discharge is likely to be more cost effective than those small devices for capturing MFs. This review demonstrates that natural fabrics would entail significantly less environmental consequences than synthetic materials. The fashion industry can also address the problem of MFs in the environment by phasing out synthetic fabrics and increase the use of natural materials.

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List of Tables

Table 1. Estimated MFs release to the aquatic environment from several example countries.

Table 2. Microfiber removal efficiency and cost of common traps and filters

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Countries	Sources	Metric tons/yr	Reference
China	Textile industry, plastic products, laundering	70,000	Statista (2019); Gavigan et al., (2020)
India	Textile industry	32,500	Statista (2019)
Japan	Textile industry	3,100	Statista (2019)
USA and Canada	Washing of all type of synthetic garments	880	Vassilenko et al., (2019)
Finland	Household washing machines	150	Sillanpää and Sainio (2017)
Norway	Laundries and households	100-600	Yang et al., (2019)

Table 2. Microfiber removal efficiency and cost of common traps and filters

Trap and filter types	Characteristics	Removal (%)	Cost (USD)	References
Guppyfriend washing bag	Polyamide washing bag of size 45 × 68 cm, with a mesh size 30 µm	39	30-35	Kärkkäinen and Sillanpää (2021)
Cora Ball - laundry ball	Soft, recycled plastic material, 13 cm diameter	10 26	40-55	Kärkkäinen and Sillanpää (2021) McIlwraith et al., (2019) Click or tap here to enter text.
Lint LUV-R filter kit	Stainless-steel mesh filter with a pore size of 150 µm in diameter	87	145	McIlwraith et al., (2019)
Filtrol™	A reusable in-line washing machine filter	89	140	Samatha et al., (2019)

List of Figures

Fig. 1. Ecological impacts of microfibers/microplastics in the aquatic environment.

Fig. 2. Factors causing the increase in microfiber release.

Fig. 3. A Sankey diagram showing the fate of microfibers especially the leakage due to low sewage coverage.

Fig. 4. Degradation time of synthetic and natural fabrics in the environment.

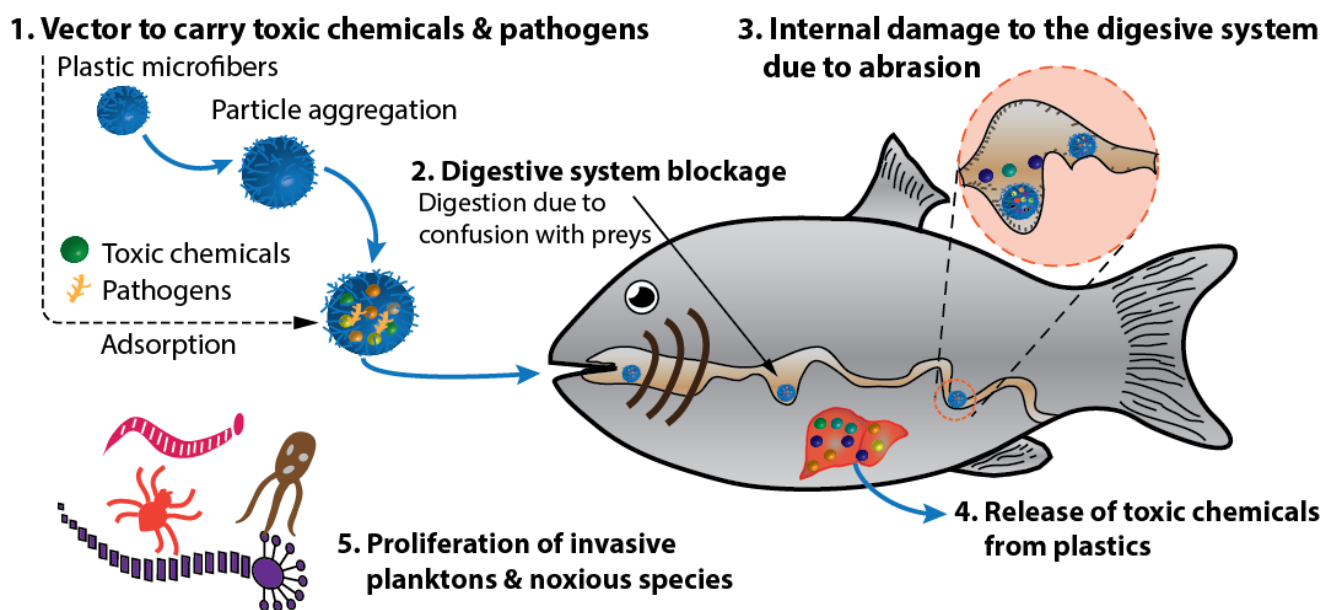


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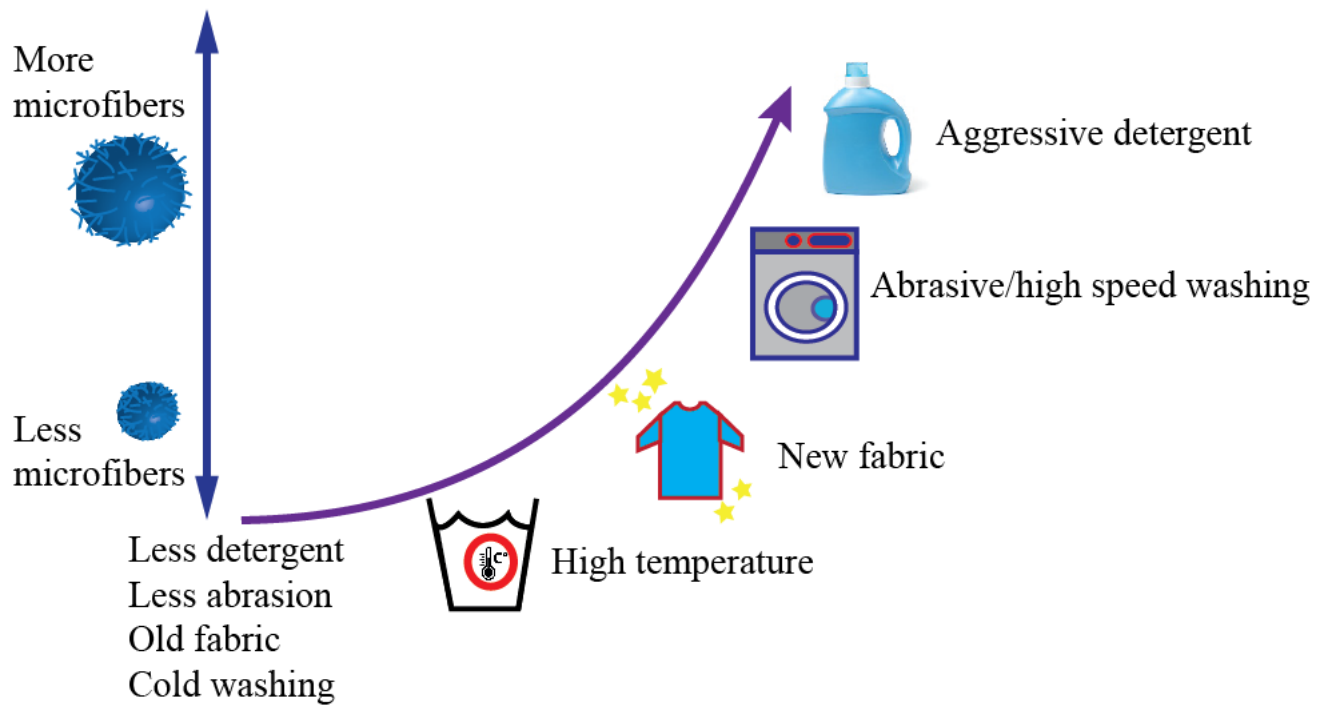


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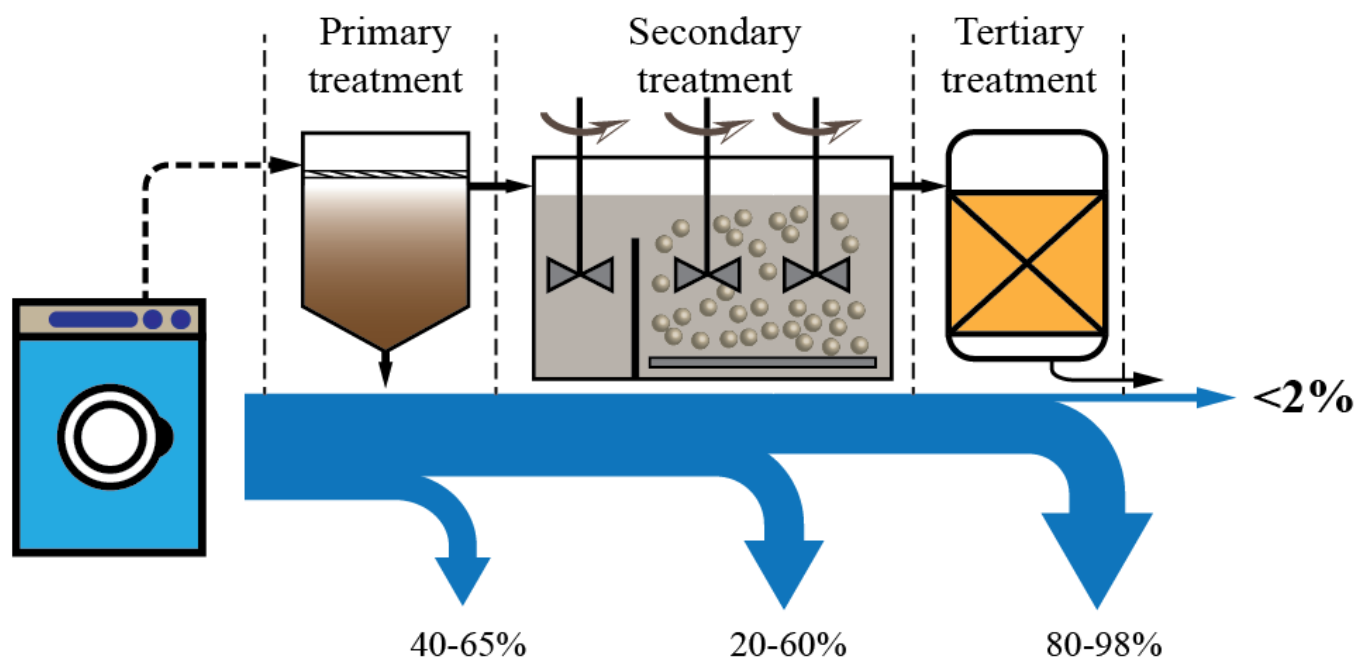


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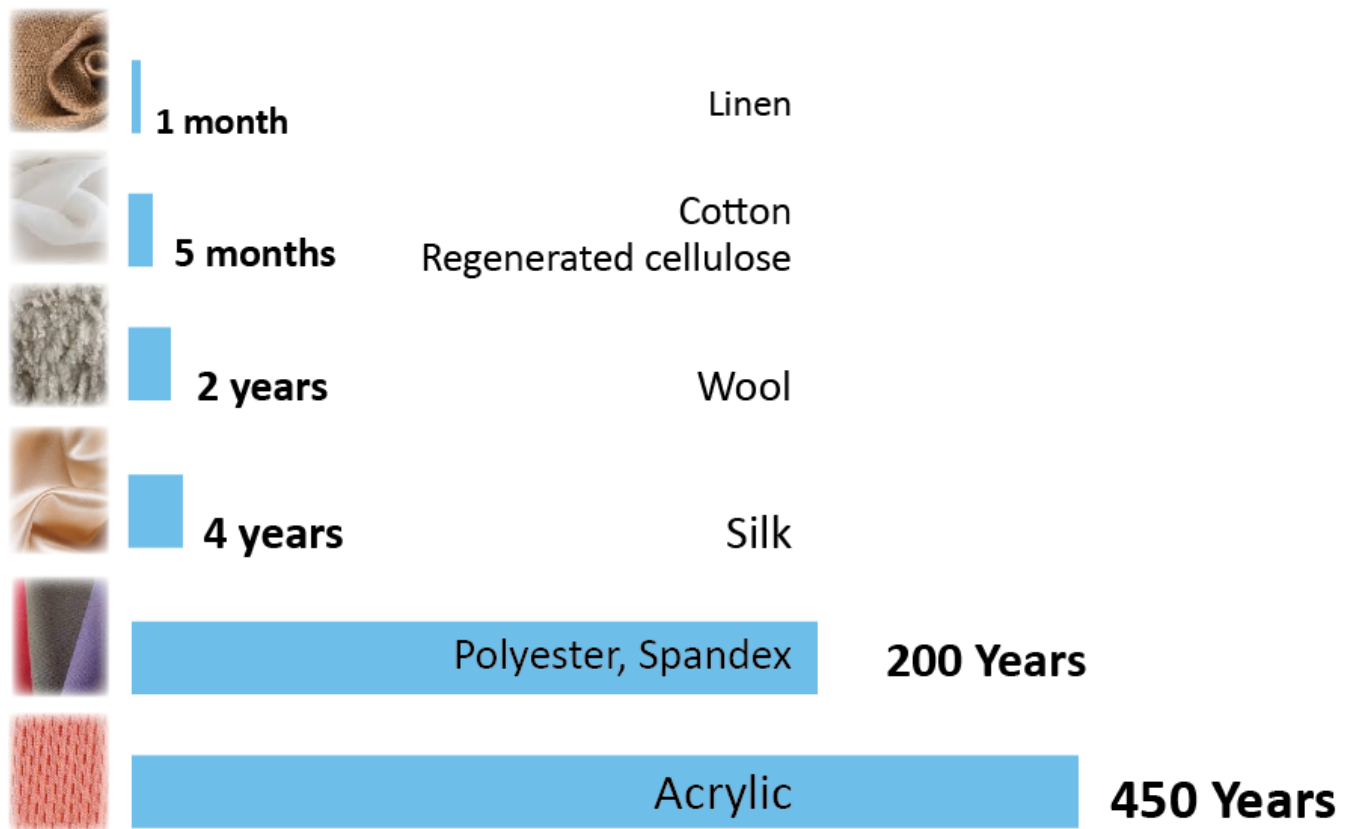


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